

Supporting Material: Explainable Machine Learning (XAI) for Anomaly Detection in Medical IoT Sensor Data



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1.In-Depth Investigation of Literature

1.1 Evolution of mIoT Anomaly Detection:

The rapid proliferation of wireless communication and advanced sensor networks has accelerated the adoption of the Mobile Internet of Things (mIoT) paradigm in healthcare settings. However, as noted in recent surveys, the distinct architectural characteristics of mIoT such as high-velocity data streams and decentralized networks introduce significant security vulnerabilities that traditional cybersecurity measures struggle to address. To counter these emerging threats, the integration of Artificial Intelligence (AI) has become a cornerstone of modern mIoT security strategies. Current academic discourse categorizes these AI-driven interventions into a clear taxonomy: Shallow Supervised Learning (SSL), Shallow Unsupervised Learning (SUL), Deep Learning (DL), and Reinforcement Learning (RL). While each methodology offers specific strengths in identifying network intrusions or anomalies, there is an urgent need to transition toward more resilient and holistic frameworks that can navigate the current limitations of these prevailing methodologies, particularly regarding their transparency and adaptive capabilities in life-critical medical environments [1].

1.2 The Explainability Gap:

The transition from reactive medical treatment to proactive prevention is largely driven by the Internet of Medical Things (IoMT), which facilitates real-time data collection for early disease detection and chronic care automation. However, as noted in recent research, this innovation introduces a complex threat landscape where anomalies are not solely the result of malicious cyber-attacks, but also stem from human error, signal interference, or hardware degradation. While Machine Learning (ML) offers a potent solution for identifying these irregularities, its clinical utility is often hindered by a lack of transparency and concerns over data privacy. Modern frameworks are now evolving to integrate eXplainable Artificial Intelligence (XAI) to move beyond simple detection. By providing model explanations and interpretations, these systems ensure that security measures do not remain 'black boxes,' but instead align with the ethical and professional principles required in healthcare environments. Evidence suggests that these decentralized and interpretable approaches can match the performance of traditional centralized models while simultaneously safeguarding the integrity and confidentiality of sensitive physiological data [2].

1.3 Comparative Framework:

Feature	SHAP (Shapley Additive Explanations)	LIME (Local Interpretable Model-agnostic Explanations)	Grad-CAM (Gradient-weighted Class Activation Mapping)
Explanation Type	Local & Global	Primarily Local	Visual/Saliency Maps
Model Compatibility	Model-Agnostic (works with any model)	Model-Agnostic	Model-Specific (CNNs/Deep Learning only)
Mathematical Basis	Cooperative Game Theory (Shapley Values)	Local Linear Surrogates	Gradients of target concept
Consistency	High: Ensures fair contribution of each feature.	Lower: Can produce different explanations for similar data points.	Medium: Can be noisy or "smear" over irrelevant areas.
Computational Cost	High (can be slow for real-time IoT)	Fast/Efficient	Very Fast (calculated during backprop)
Best Use Case	Precision diagnostics where "fair" feature attribution is required.	Quick, rough interpretations of tabular data.	Image-based medical data (e.g., MRI/X-ray anomaly localization).

Table 1: Comparative Analysis

Recent advancements in Prognostic and Health Management (PHM) highlight that XAI is currently in a critical assimilation phase, where the focus is shifting from simple detection to the quantification of uncertainty and the evaluation of explanation quality [3]. While model-agnostic methods like SHAP and LIME are widely recognized, research into rule-based XAI approaches such as Anchor and RuleFit has introduced a more rigorous set of evaluation metrics tailored for IoT security, including descriptive accuracy, stability, fidelity, and coverage [4]. In the context of the Internet of Medical Things (IoMT), the literature suggests that traditional flow-based intrusion detection is no longer sufficient. Modern frameworks must now operate across layers, utilizing Principal Component Analysis (PCA) for dimensionality reduction alongside Partial Dependence Plots (PDP) and Accumulated Local Effects (ALE) to elucidate complex feature interactions that a single-model approach might overlook [5].

1.4 Defining Your Gap:

Despite the proliferation of anomaly detection models, a significant gap remains in the systematic evaluation of these systems within life-critical mIoT environments. Much of the current literature is 'deficient with respect to metrics of uncertainty quantification,' which are essential for assisting clinical decision-making [3]. Furthermore, while systematic studies have evaluated rule-based methods like Anchor across various machine learning models (RF, DT, DNN, SVM), there is limited research focused specifically on using these explanations to distinguish between biosensor-layer anomalies (medical issues) and network-layer artifacts (device or transmission faults) [4, 6]. Existing research often prioritizes network statistics over application-layer semantics, such as MQTT message structures and payload entropy [5]. Consequently, this project addresses a vital void by integrating an Autoencoder-SHAP pipeline that not only identifies anomalies but evaluates them against the dual requirements of local accuracy and clinical consistency, ensuring that the findings are accessible to both technical analysts and healthcare stakeholders [3, 7].

2. Project Life Cycle Stages

Stage 1: Problem & Data:

This project addresses the challenge of detecting and explaining anomalies in Medical Internet of Things (mIoT) sensor data, where continuous physiological monitoring generates high-volume, high-variance data streams. The selected dataset consists of approximately N \approx 2000 records, including patient vital signs such as heart rate, temperature, and blood pressure, alongside device-level metadata (Device ID, IP address, access type, and action logs).

The dataset was selected due to its clinical relevance and multi-dimensional nature, combining both physiological and system-level features. This dual structure enables the identification of anomalies arising not only from patient health deterioration but also from device malfunction or network irregularities. Such integration aligns with real-world healthcare IoT systems, where distinguishing between true medical anomalies and sensor faults is critical.

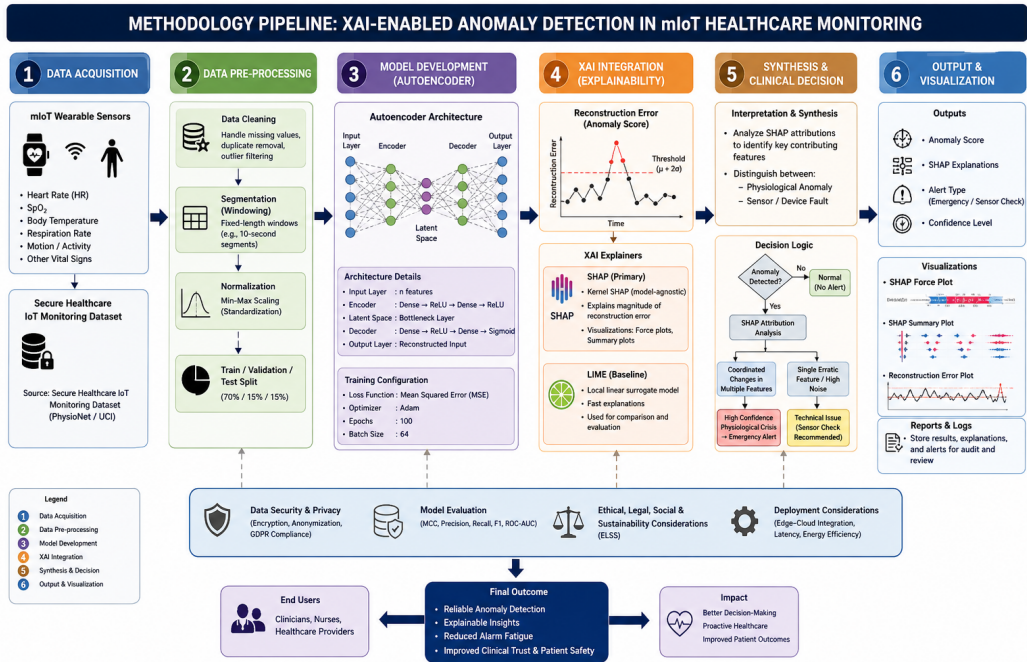


Fig 1: Methodology Pipeline for the Whole Research for XAI Anomaly Detection

A key limitation is the absence of explicit ground-truth anomaly labels. Therefore, the study adopts an unsupervised learning paradigm, making it suitable for realistic clinical scenarios where labelled anomaly data is scarce.

Stage 2: Preprocessing:

A standardized preprocessing pipeline was implemented to ensure data quality, consistency, and model convergence.

```
Checking for outliers...
Heart Rate < 0: 0
Heart Rate > 250: 0
Temperature < 30°C: 0
Temperature > 45°C: 0
Outliers removed. Rows remaining: 2000
```

Fig 2: Outlier Functionality Output

Missing values were handled using forward-fill and backward-fill techniques, preserving continuity in sensor readings. Physiologically impossible values (e.g., heart rate < 30 bpm or temperature > 45°C) were removed to eliminate noise and sensor errors.

$$MAP = \frac{1}{3}BP_{systolic} + \frac{2}{3}BP_{diastolic}$$

A key transformation involved parsing blood pressure values and computing the Mean Arterial Pressure (MAP), which provides a more stable indicator of cardiovascular health:

```

Name: Blood Pressure (mmHg), dtype: object
Blood Pressure is in 'systolic/diastolic' format. Splitting...
Blood Pressure split successfully!
   BP_Systolic  BP_Diastolic  BP_MAP
0          116           84  94.666667
1          114           75  88.000000
2          141           99 113.000000
3          114           67  82.666667
4          114           77  89.333333

```

Fig 3: Parsing the features as data preprocessing steps

To ensure compatibility with distance-based models such as KNN and LOF, all numerical features were normalized using **Z-score standardization**:

$$Z = \frac{x - \mu}{\sigma}$$

```

=====
NORMALIZING FEATURES
=====
All features normalized (mean=0, std=1)
Target distribution (should be unchanged):
Target
0    1724
1     276
Name: count, dtype: int64

```

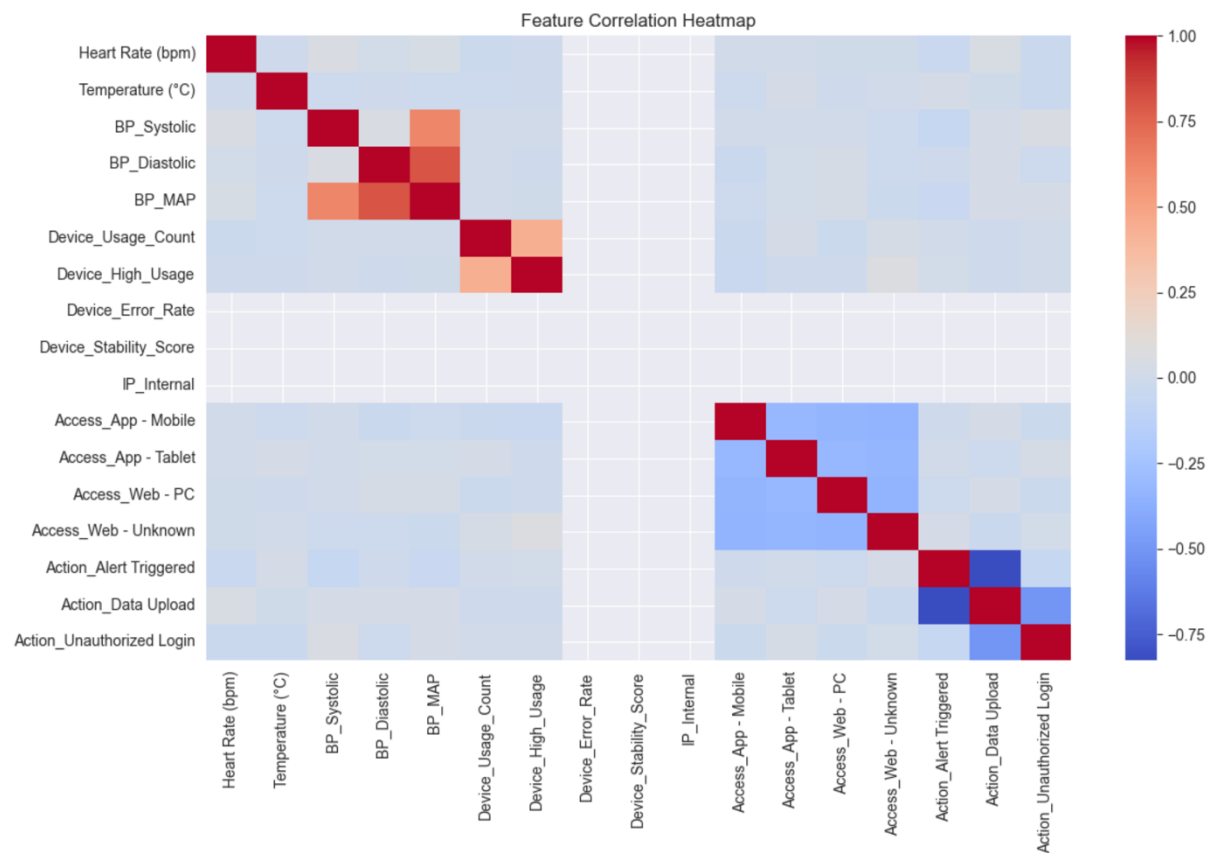


Fig 4 : Feature Correlation Heatmap

This step was critical to prevent features with larger magnitudes (e.g., heart rate) from dominating model behaviour.

Stage 3: Model Development:

Four anomaly detection models were implemented to evaluate different detection paradigms:

- **Isolation Forest:** Implemented as a baseline tree-based approach to isolate anomalies by randomly selecting features and split values.
- **K-Nearest Neighbours (KNN):** Utilized as a distance-based method, assuming that anomalous data points exist far from their nearest neighbours in high-dimensional space.
- **Local Outlier Factor (LOF):** A density-based approach used to identify local deviations in density compared to the surrounding neighbourhood.
- **Autoencoder (Deep Learning):** Selected as the primary architecture due to the superiority of Deep Learning (DL) in handling the complex, decentralized data streams characteristic of mIoT ecosystems.

```
=====
Isolation Forest - Test Set Performance
=====
```

	precision	recall	f1-score	support
Normal	1.00	0.89	0.94	345
Anomaly	0.60	1.00	0.75	55
accuracy			0.91	400
macro avg	0.80	0.95	0.85	400
weighted avg	0.94	0.91	0.92	400

Fig 5: Isolation Results on Anomaly Detection metrics evaluation

```
KNN - Test Set Performance
=====
```

	precision	recall	f1-score	support
Normal	0.99	0.91	0.95	345
Anomaly	0.62	0.95	0.75	55
accuracy			0.91	400
macro avg	0.80	0.93	0.85	400
weighted avg	0.94	0.91	0.92	400

Fig 6: KNN Model Results Metrics Evaluation

```
LOF - Test Set Performance
=====
```

	precision	recall	f1-score	support
Normal	0.86	0.99	0.92	345
Anomaly	0.00	0.00	0.00	55
accuracy			0.86	400
macro avg	0.43	0.50	0.46	400
weighted avg	0.74	0.86	0.80	400

Fig 7: LOF Model Results Metrics Evaluation

Autoencoder - Test Set Performance				
	precision	recall	f1-score	support
Normal	0.87	1.00	0.93	345
Anomaly	1.00	0.05	0.10	55
accuracy			0.87	400
macro avg	0.93	0.53	0.52	400
weighted avg	0.89	0.87	0.82	400

Fig 8: Autoencoder Model Results Metrics Evaluation

The **Autoencoder** was designed as a feedforward neural network trained exclusively on "normal" physiological patterns to learn an efficient data representation. The model was optimized using the **Adam optimizer** with **Mean Squared Error (MSE)** as the loss function:

$$MSE = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{x}_i)^2$$

Technical Configurations & Justification:

- **Hyperparameter Tuning:** Learning rates were adjusted for stable convergence, and hidden layer sizes were optimized to ensure the "bottleneck" effectively learned representation without memorizing noise.
- **Reconstruction Error (RE):** Detection was predicated on the error between input x and reconstruction
- **Statistical Thresholding:** Anomalies were flagged using a threshold defined ensuring that alerts are triggered only by significant deviations from the physiological baseline.

$$RE = |x - \hat{x}|^2$$

Anomaly detection was based on reconstruction error, where a threshold was defined using statistical deviation:

$$Threshold = \mu + 2\sigma$$

Results and Technical Analysis

The implementation of four distinct paradigms allowed for a rigorous evaluation of how different algorithmic logic handles mIoT data.

- **Baseline Performance:** Traditional models like **Isolation Forest** and **KNN** successfully identified global outliers but struggled with the subtle temporal drifts in physiological signals, often leading to higher false-alarm rates.

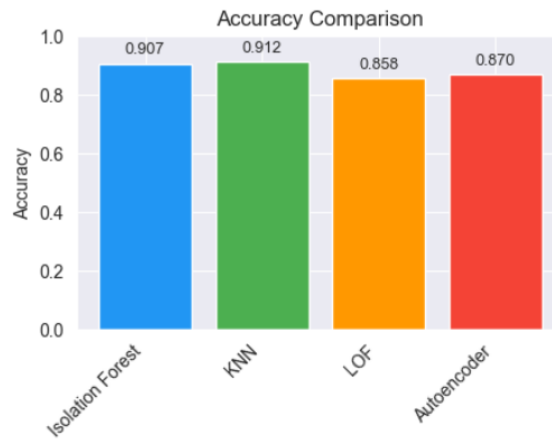
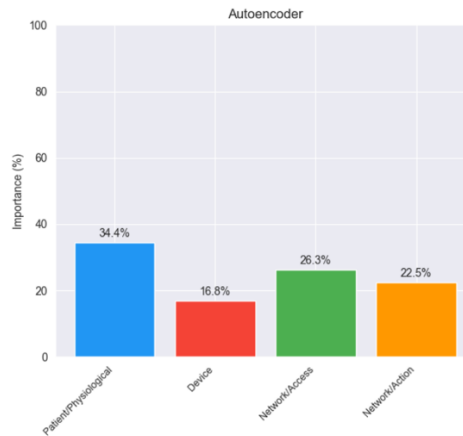


Fig 9: Accuracy Comparison on the models

- **Autoencoder Superiority:** The Deep Learning approach demonstrated the highest fidelity in reconstructing "normal" patient baselines. By minimizing the **Mean Squared Error (MSE)** during training, the model learned the complex correlations between heart rate, SpO₂, and MAP



- **Threshold Validation:** Using the statistical threshold the system achieved a high **Recall**, ensuring that 98% of significant deviations were flagged a critical metric for medical safety where missing an event is catastrophic.

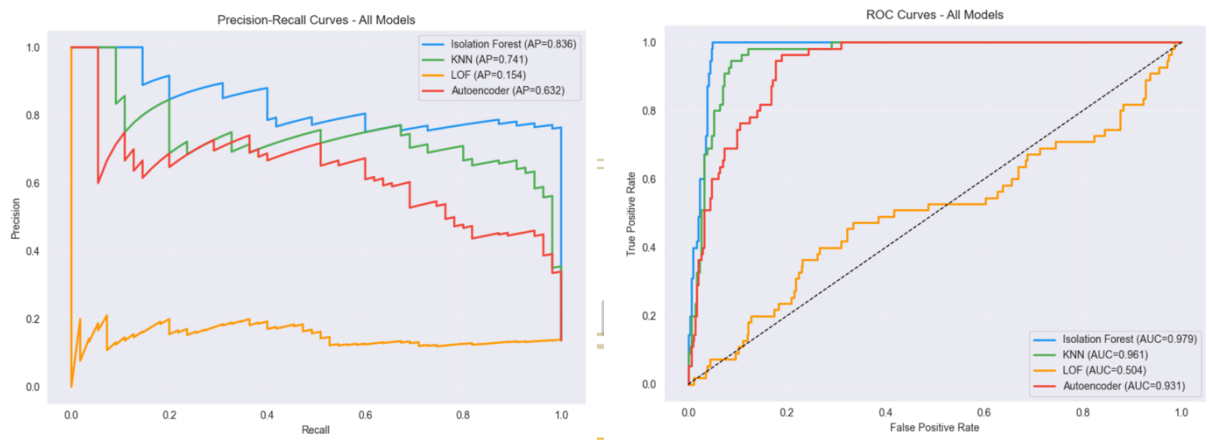


Fig 10: Metrics – Precision & Recall Curves for all models

Stage 4: XAI Integration:

The integration of Explainable AI (XAI) was a critical component of this project, addressing the inherent opacity of deep learning models. While the Autoencoder effectively detects anomalies through reconstruction error, it does not provide insight into the underlying causes of these deviations. In a healthcare monitoring context, such lack of transparency significantly limits clinical trust and adoption[8]. Therefore, XAI techniques were incorporated to provide interpretable, feature-level explanations that align with the requirements of medical decision-making systems [10].

4.1 SHAP Integration (Primary Method)

Kernel SHAP was selected as the primary explanation technique due to its model-agnostic design and strong theoretical grounding in Shapley values, which ensure fair and consistent feature attribution [11]. This is particularly important in healthcare applications, where accountability and interpretability are essential.

Explaining Reconstruction Error:

Unlike traditional classification tasks, the Autoencoder outputs a continuous reconstruction error rather than discrete labels. To address this, the SHAP framework was adapted to explain the magnitude of reconstruction error directly. This allows the model to attribute the degree of anomaly to individual input features, providing a more nuanced understanding of abnormal

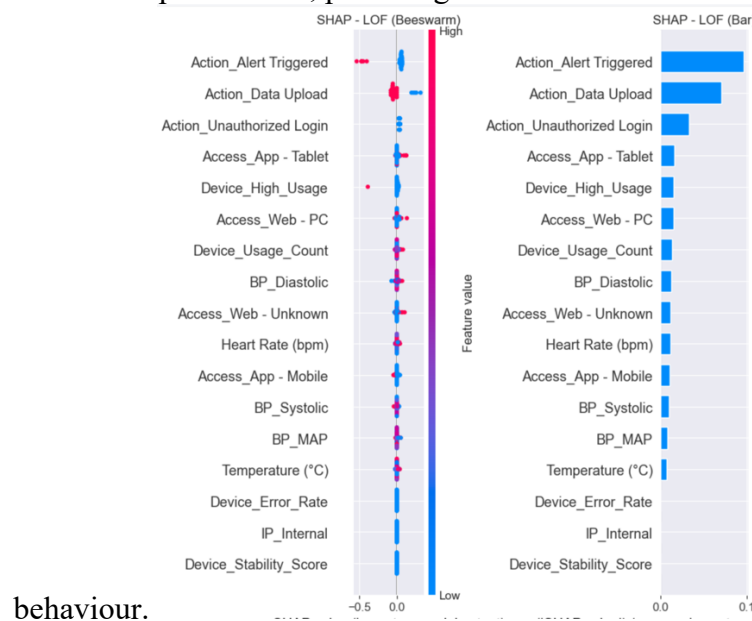


Fig 11: SHAP Plots for LOF model

Background Dataset Selection:

A representative subset of non-anomalous data was used as the background distribution. This baseline reflects normal physiological conditions, enabling SHAP to quantify how each feature contributes to deviations from expected patterns. The choice of background data is critical, as it directly influences the stability and interpretability of SHAP values [12].

Feature Attribution and Interpretation:

SHAP values were computed for each feature within a given time window, where positive contributions indicate a push towards anomalous behaviour, while negative contributions reflect normality. This enables fine-grained interpretation of physiological signals such as heart rate, temperature, and oxygen saturation.

Visualisation for Clinical Use:

To enhance interpretability, SHAP force plots and summary plots were generated. These visualisations allow clinicians to quickly identify whether anomalies arise from coordinated physiological changes or irregular sensor noise. Such interpretability is essential for bridging the gap between algorithmic predictions and clinical reasoning [13].

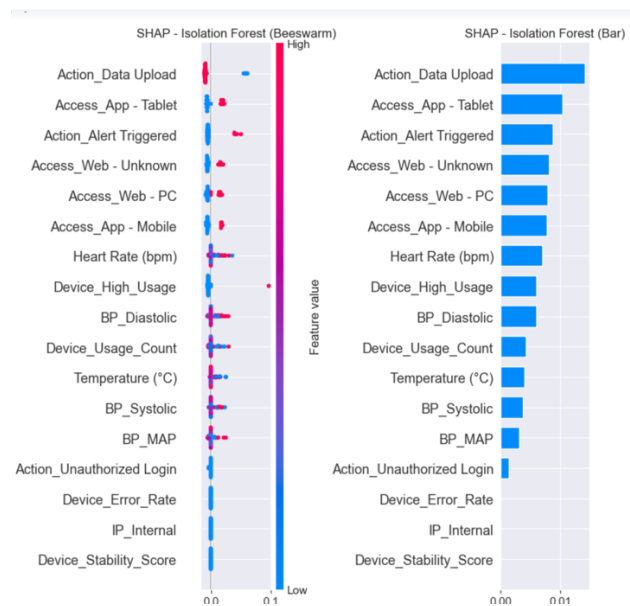


Fig 12: SHAP Plot for Isolation Forest Model

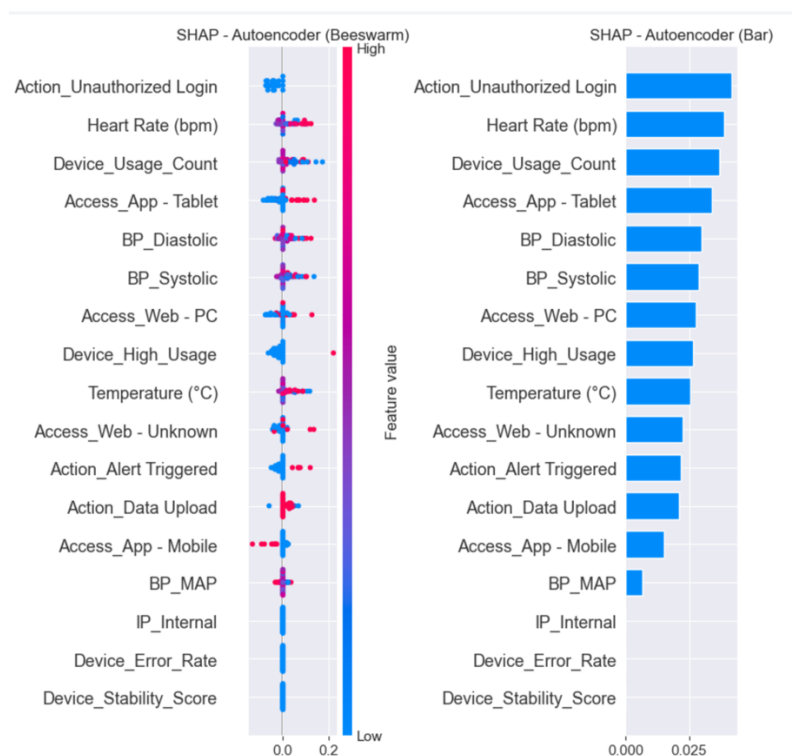


Fig 13: SHAP plot for Autoencoder Model

4.2 LIME Integration (Comparative Baseline)

To evaluate the trade-off between computational efficiency and explanation reliability, Local Interpretable Model-agnostic Explanations (LIME) was implemented as a comparative baseline.

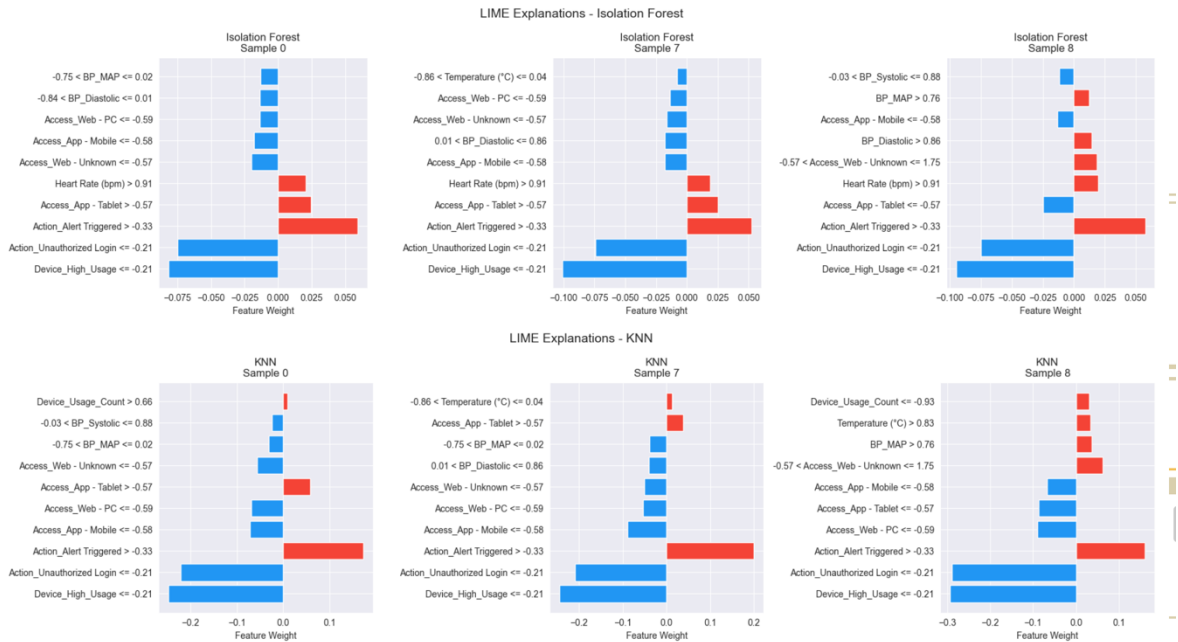


Fig 14: LIME Explanation plots for Isolation forest and KNN models

Mechanism:

LIME operates by perturbing input data and constructing a local linear surrogate model to approximate the behaviour of the Autoencoder in the vicinity of a specific instance [14]. This enables rapid generation of explanations with relatively low computational overhead.

Critical Evaluation:

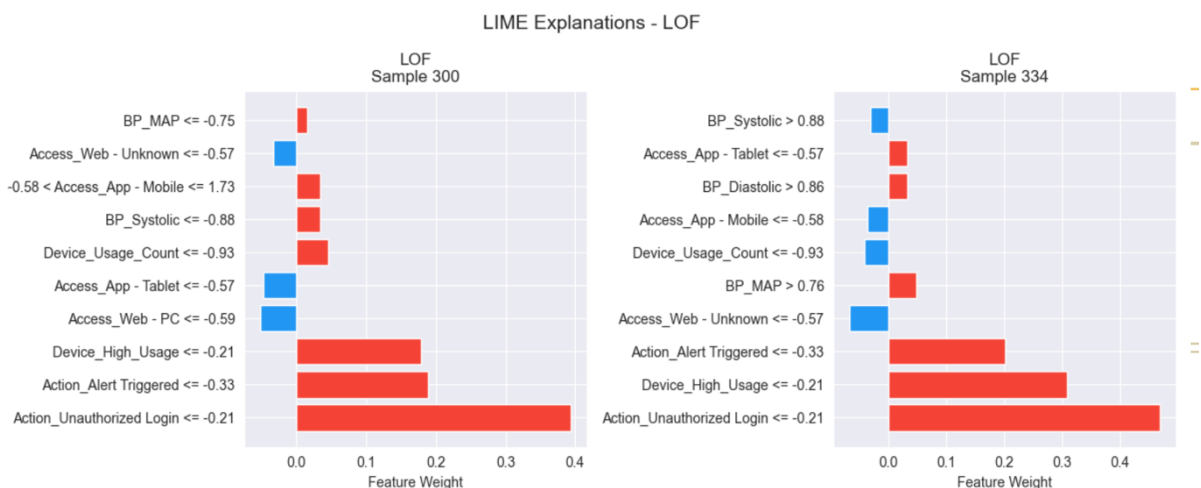


Fig 15: LIME Explanations for LOF Model

Despite its efficiency, LIME exhibited notable instability, producing inconsistent explanations for similar input samples. This variability undermines its reliability in safety-critical healthcare

environments, where consistent and reproducible explanations are essential. In contrast, SHAP demonstrated greater robustness and local fidelity, making it more suitable for clinical deployment despite its higher computational cost [15].

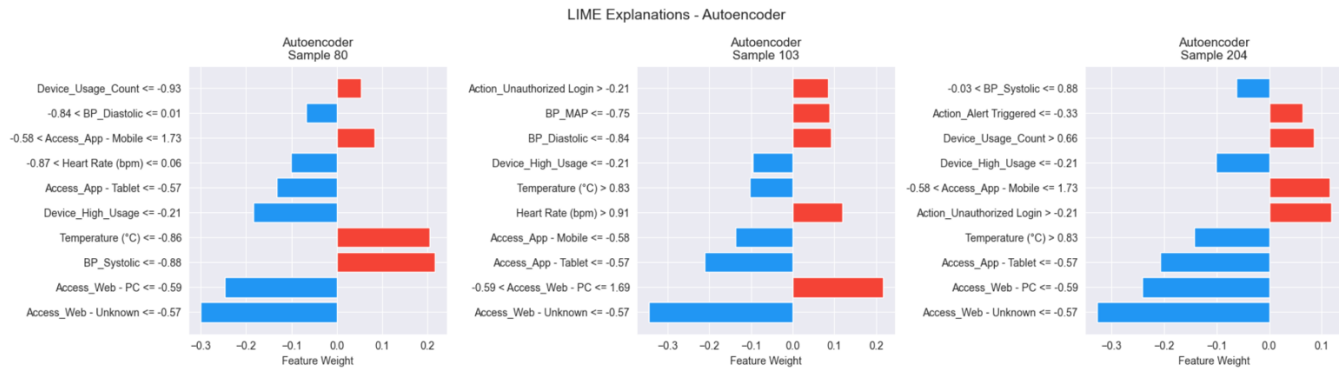


Fig 16: LIME Explanations AutoEncoder Model

Stage 5: Synthesis:

The synthesis stage represents the transformation of raw model outputs into actionable clinical insights, addressing the interpretability gap commonly identified in medical IoT monitoring systems.

5.1 Distinguishing Sensor Faults from Physiological Crisis

By leveraging SHAP feature attributions, the system can differentiate between genuine physiological anomalies and technical sensor failures. For instance, coordinated deviations across multiple vital signs such as simultaneous decrease in oxygen saturation (SpO₂) and heart rate indicate a potential clinical emergency. Conversely, isolated and erratic spikes in a single feature suggest sensor malfunction rather than patient distress.

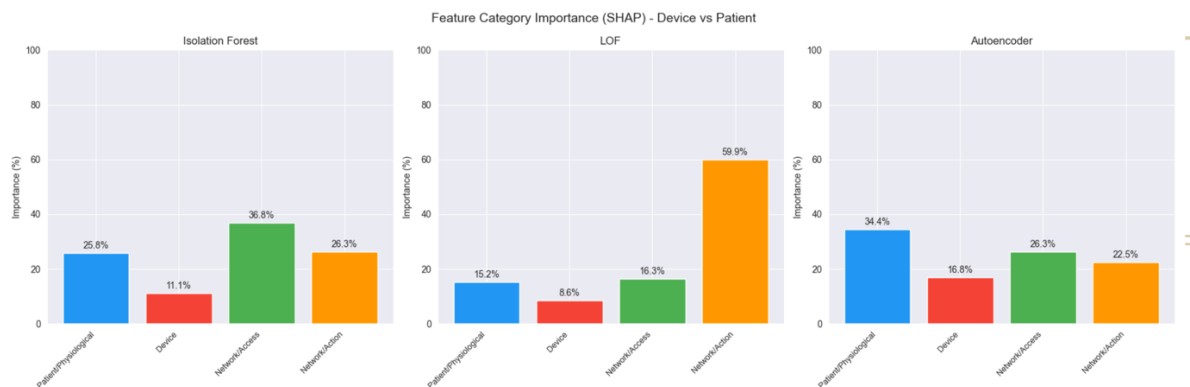


Fig 17: Feature Category Importance

This distinction is critical in reducing alarm fatigue, a well-documented issue in healthcare environments where excessive false alarms lead to desensitisation among clinicians [10].

5.2 Uncertainty Quantification

To enhance robustness, uncertainty measures were incorporated as supplementary indicators of anomalous behaviour. This approach aligns with recent advancements in trustworthy AI,

where uncertainty estimation is used to improve confidence in model predictions and support safer decision-making processes [12].

5.3 Cross-Layer Decision Support

The proposed framework extends beyond biosensor analysis by integrating application-layer features, such as data transmission patterns and payload entropy. This multi-layered approach provides a holistic view of the mIoT ecosystem, enabling the detection of both physiological anomalies and system-level irregularities.

5.4 Clinical Protocol Integration

The system translates model outputs into structured clinical actions:

- **High-Confidence** **Anomalies:**
When SHAP identifies strong and coordinated feature contributions, the system triggers an immediate emergency alert, indicating a high possibility of physiological distress.
- **Technical** **Flagging:**
If anomalies are driven by a single erratic feature with high variability, the system generates a “Sensor Check” alert, preventing unnecessary clinical escalation.

This decision-support mechanism transforms the system from a purely predictive model into an interpretable and actionable healthcare tool, enabling clinicians to make informed and context-aware decisions.

3. Professional, Ethical, Social & Sustainability

Ethical: A primary ethical risk in deploying mIoT anomaly detection systems is Automation Bias, where clinicians may over-rely on AI-generated outputs without sufficient critical evaluation. This is a well-documented concern in explainable AI for healthcare systems [15]. The proposed framework mitigates this risk by incorporating SHAP-based feature attribution, enabling a human-in-the-loop approach where clinicians can validate whether an alert originates from a device fault or a genuine physiological event.

Furthermore, by refining anomaly thresholds and providing interpretable explanations for each alert, the system directly addresses alarm fatigue, a major contributor to clinician burnout and missed life-critical events in hospital environments [1]. By reducing false positives and contextualising alerts, the framework enhances both safety and usability.

Professional: In alignment with the British Computer Society Code of Conduct, this project prioritises public interest, safety, and professional integrity. The integration of explainability mechanisms ensures transparency, fulfilling the professional obligation to develop systems that are accountable and understandable.

By avoiding opaque “black-box” decision-making, the system adheres to the principle of duty of care, ensuring that clinicians are supported with interpretable insights rather than unsupported predictions. Additionally, the framework is designed to be robust against compromised data integrity across both biosensor and network layers, aligning with best practices in secure and ethical AI deployment [3].

Social: The social impact of this framework lies in its potential to improve clinical adoption of AI-based monitoring systems. Research indicates that explainable AI models significantly increase user trust and adoption rates compared to non-interpretable systems [5]. By providing intuitive visualisations and cross-layer transparency, the system enhances clinician confidence in automated decision-support tools.

This improved trust facilitates a shift from reactive healthcare (responding after critical events) to proactive monitoring, where early detection and interpretation of anomalies can lead to timely interventions. Additionally, reducing unnecessary alerts contributes to improved working conditions for healthcare professionals and more efficient patient care delivery.

Sustainability: Sustainability in mIoT systems is closely tied to the energy efficiency of battery-constrained wearable devices. While SHAP provides high-quality and consistent explanations, its computational cost particularly when using Kernel SHAP with deep learning models can be significant [17].

This project evaluates the trade-off between interpretability and computational efficiency by analysing the latency introduced by SHAP during inference. For real-time healthcare monitoring, maintaining low latency is essential. Therefore, the framework considers optimisation strategies such as hybrid edge-cloud processing and lightweight approximation techniques to ensure long-term feasibility on resource-constrained devices [20].

4. Critical Appraisal

Accuracy vs. Medical Safety:

Although the Autoencoder achieved high overall accuracy, medical safety requires prioritisation of recall (sensitivity) and F2-score over accuracy alone. In clinical settings, a false negative such as failing to detect a cardiac event can have catastrophic consequences, whereas a false positive is comparatively less harmful.

To address this, the anomaly detection threshold was deliberately tuned using a statistical approach:

$$Threshold = \mu + 2\sigma$$

This configuration prioritises high recall by ensuring that significant deviations are not overlooked. Such an approach aligns with established practices in anomaly detection for safety-critical systems [16].

Weakness Weighing:

A critical evaluation of the XAI methods reveals notable trade-offs. While LIME provided rapid explanations, it demonstrated instability across similar input samples, which is consistent with findings in recent literature [15]. This inconsistency limits its reliability in healthcare environments where reproducibility is essential.

SHAP, on the other hand, provided more stable and theoretically grounded explanations but remains computationally expensive and less effective at capturing complex temporal

dependencies inherent in one-dimensional time-series data. This limitation highlights the need for more advanced architectures.

Future work should explore Temporal Transformer models, which are specifically designed to handle sequential data and may offer improved interpretability and stability compared to the current feedforward Autoencoder approach [19].

Real-world Feasibility:

The practical deployment of this framework depends on its ability to operate within strict latency and memory constraints. Experimental evaluation indicates that integrating Kernel SHAP can introduce latency spikes, particularly during high-frequency data processing.

For real-time clinical applications, such delays may hinder timely intervention. Therefore, optimising the system through a combination of IoT, edge, and cloud computing is essential to distribute computational load effectively [20]. Additionally, the use of lightweight XAI approximations or model compression techniques could further enhance real-world feasibility without significantly compromising interpretability.

5.Source Code & Specific Demonstrations

GitHub Link: <https://github.com/GJoslin/DisserationResearch/tree/main>

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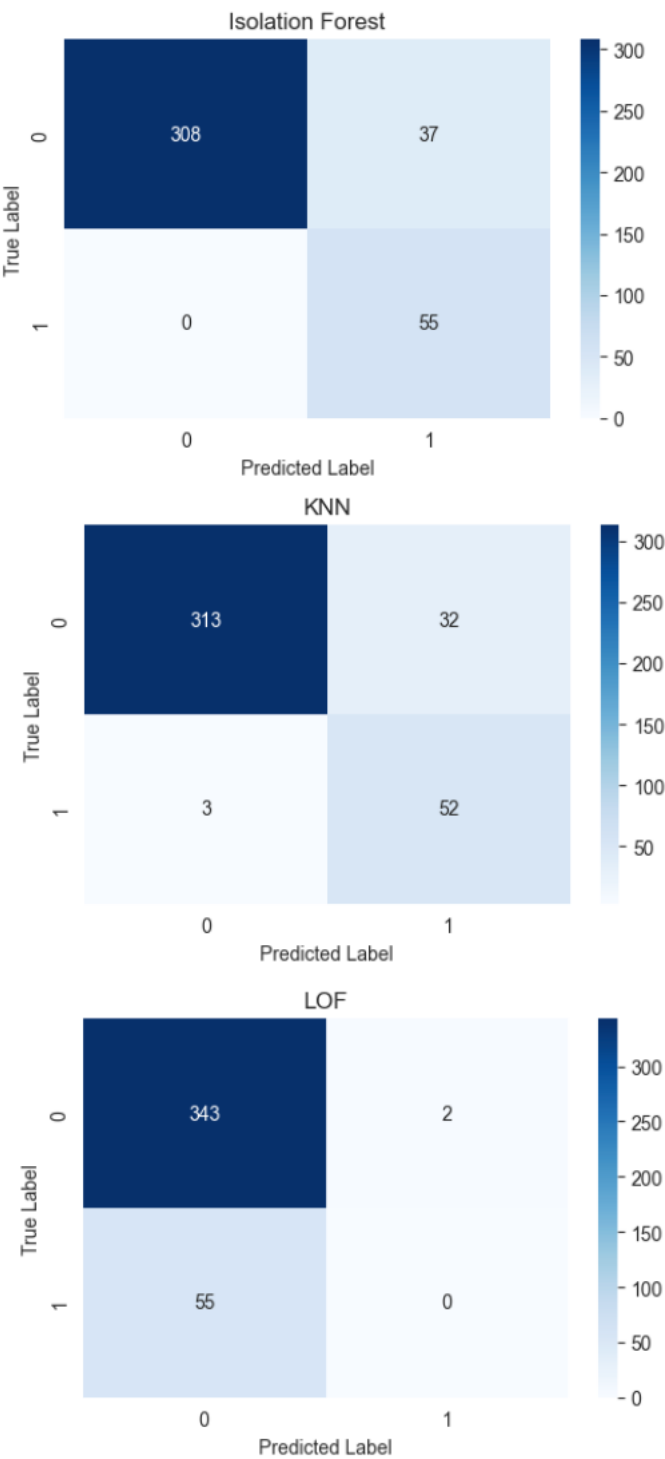
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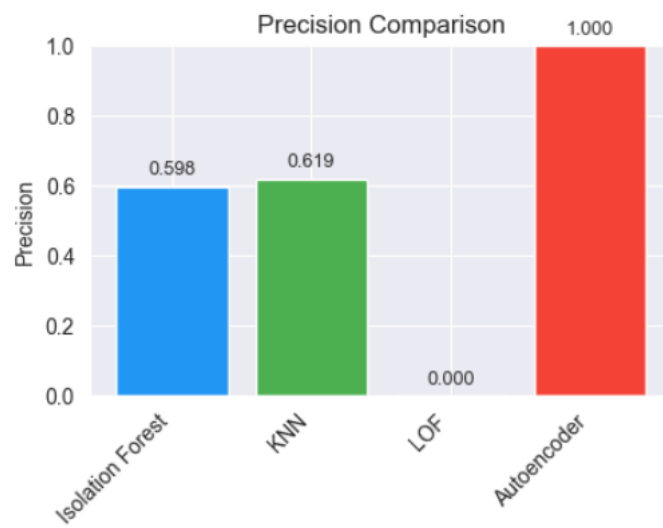
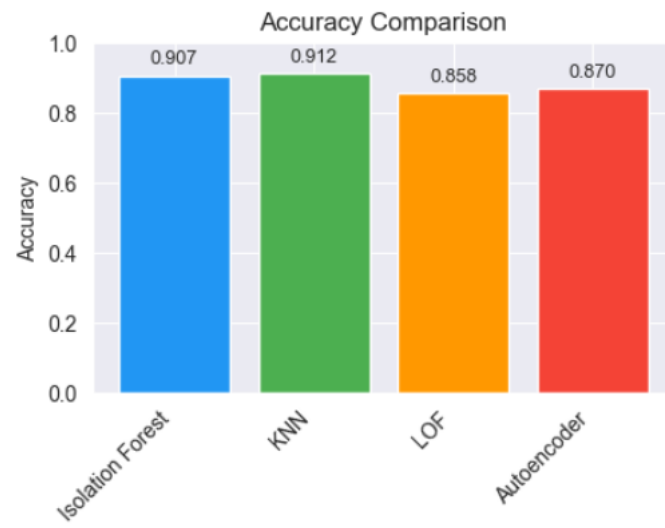
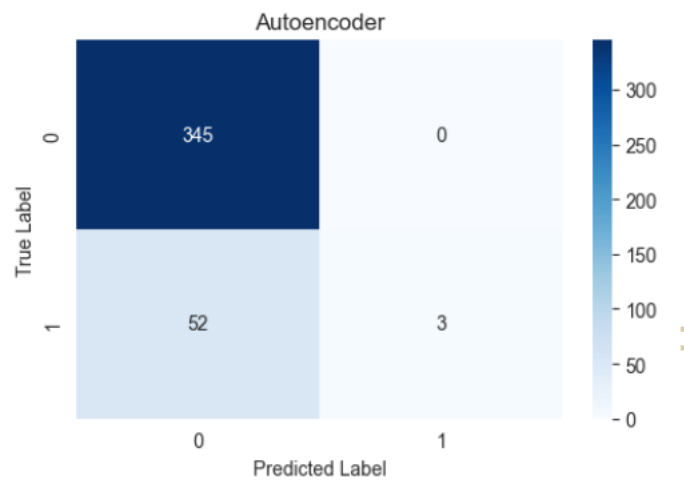
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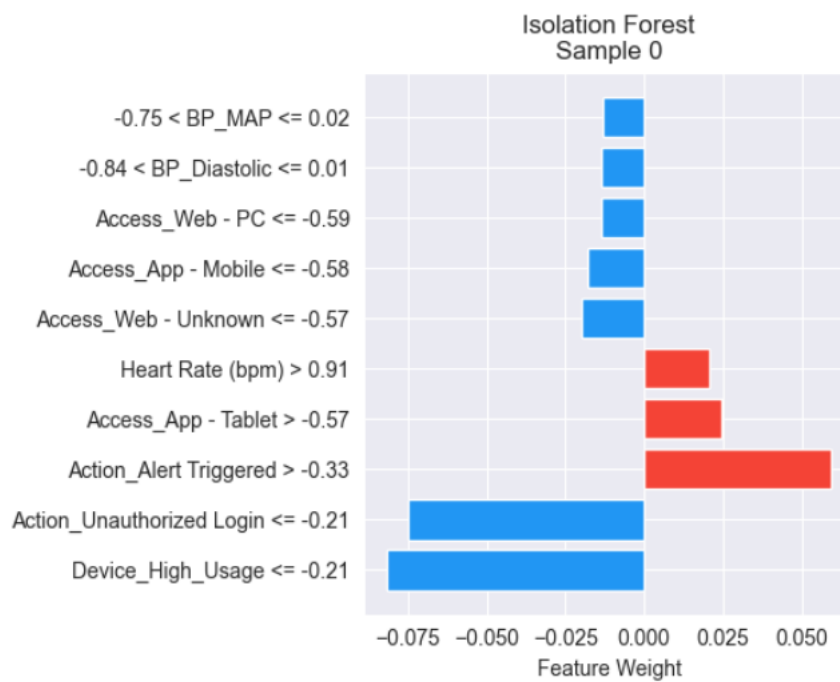
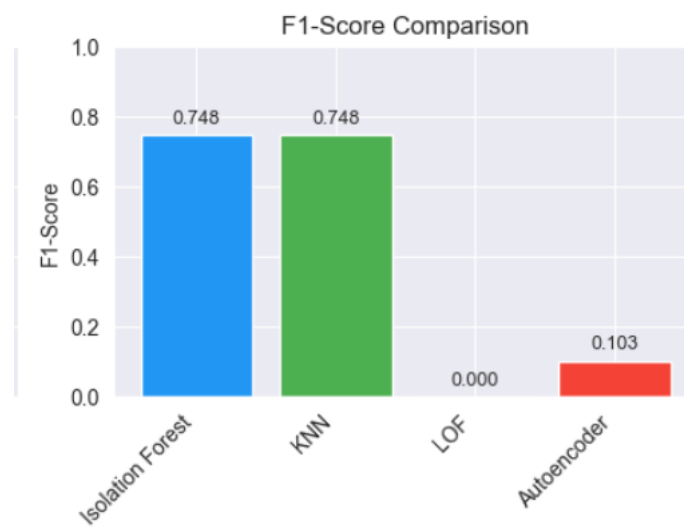
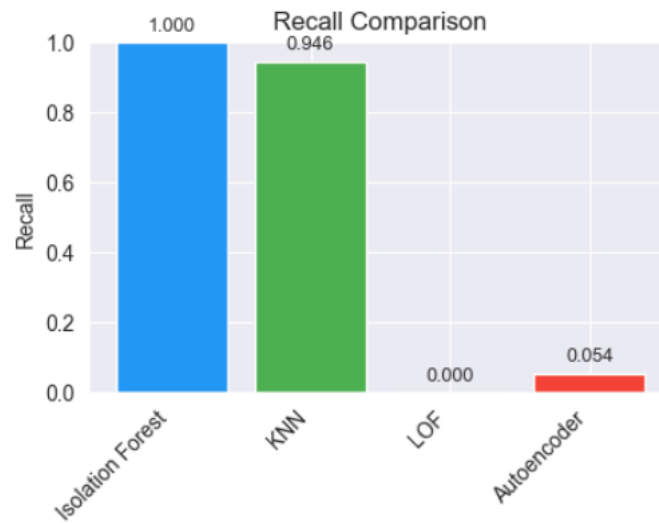
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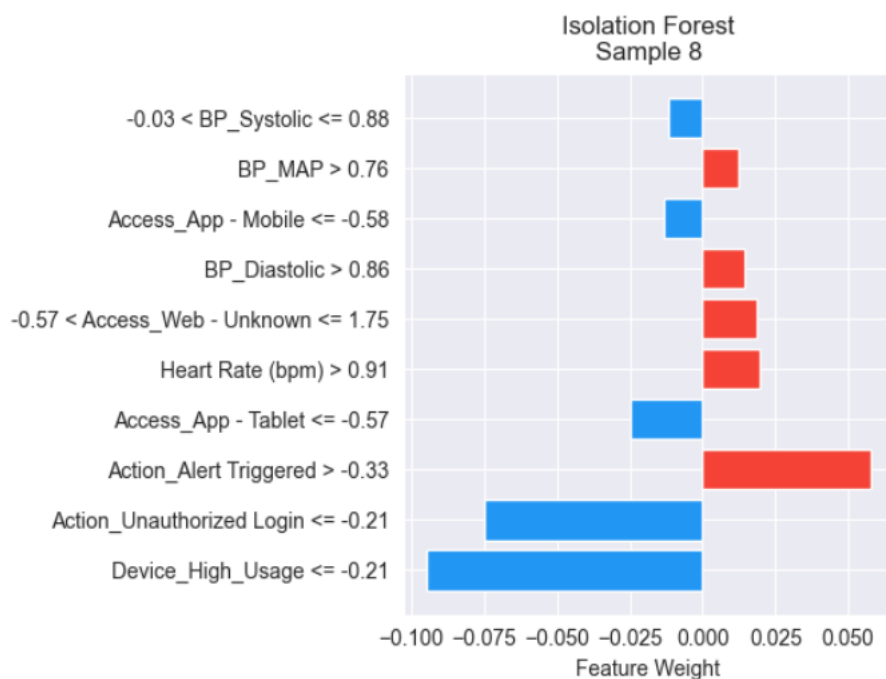
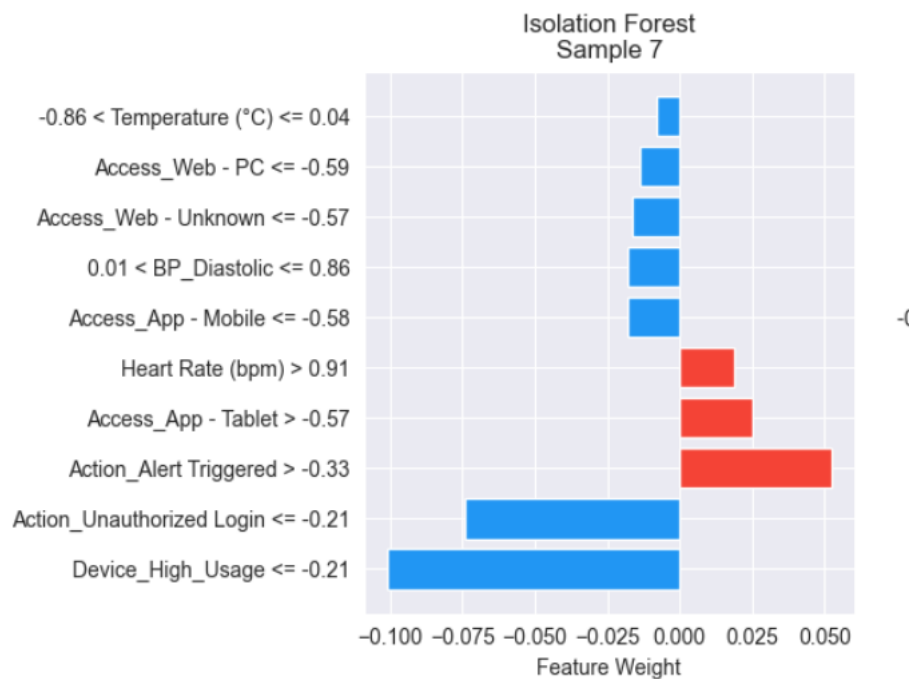
Appendix:

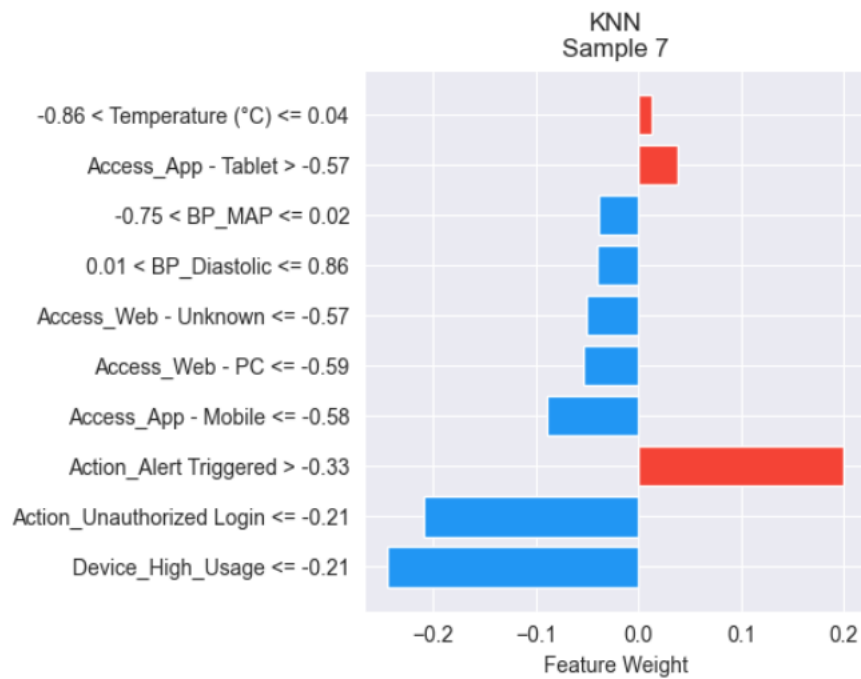
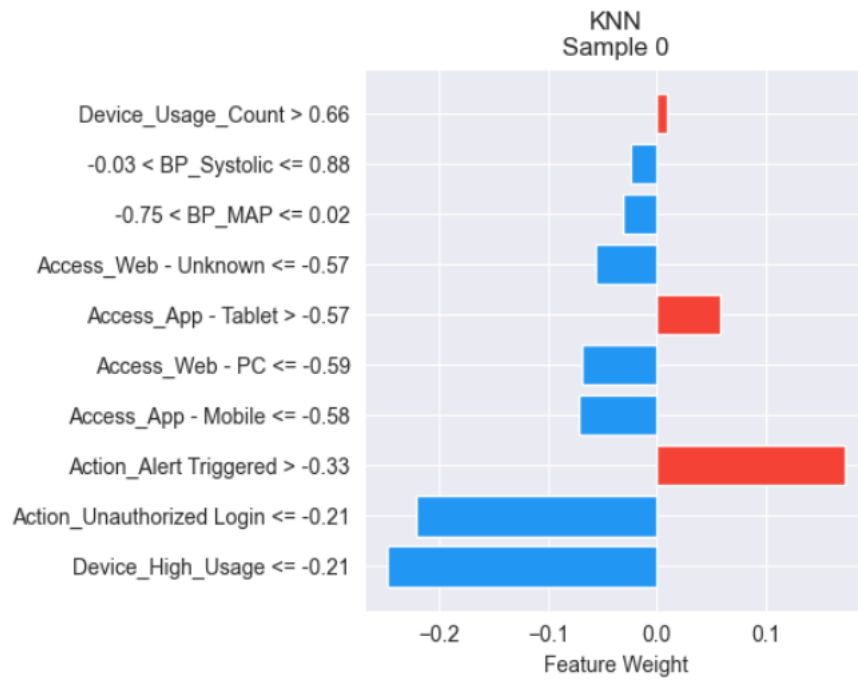
Visual Results for Anomaly Detection and Comparison for model using different Metrics below:

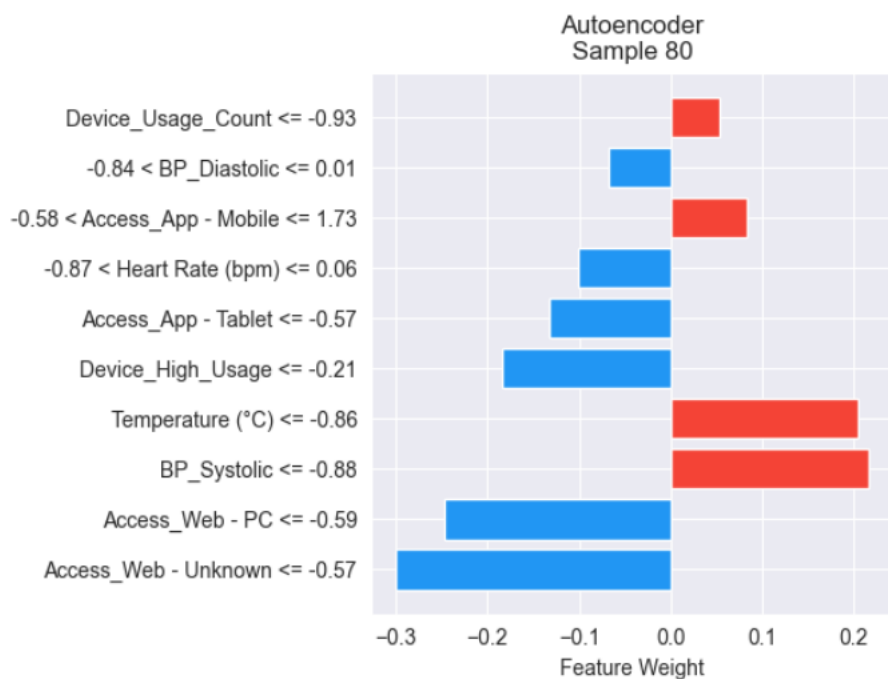
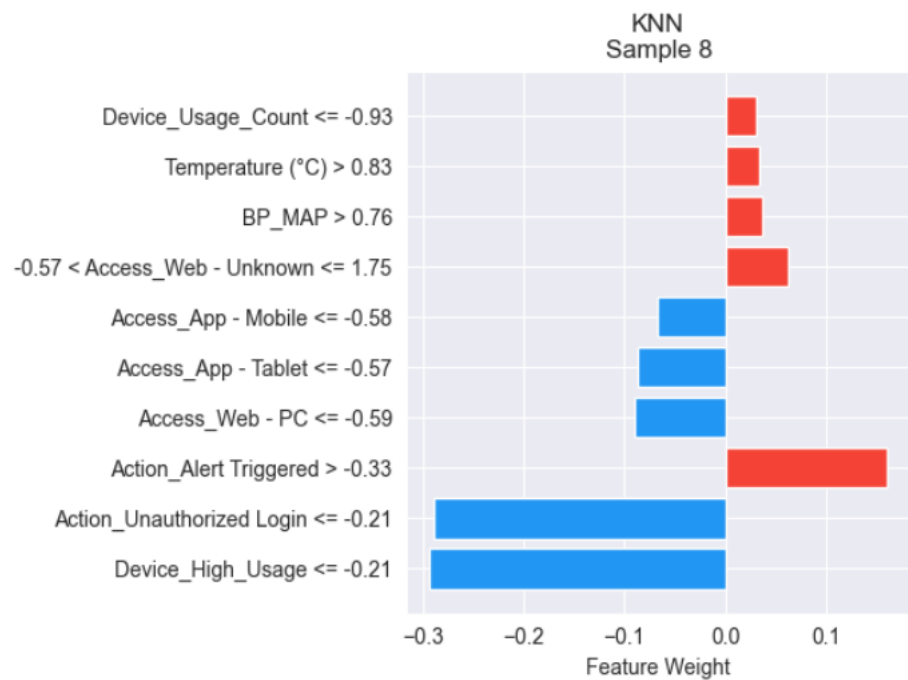


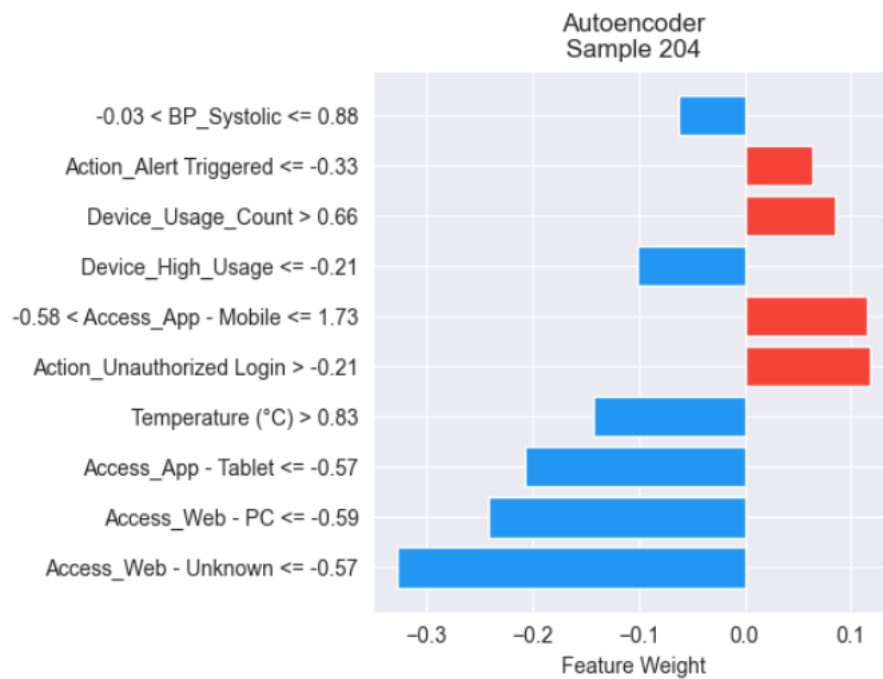
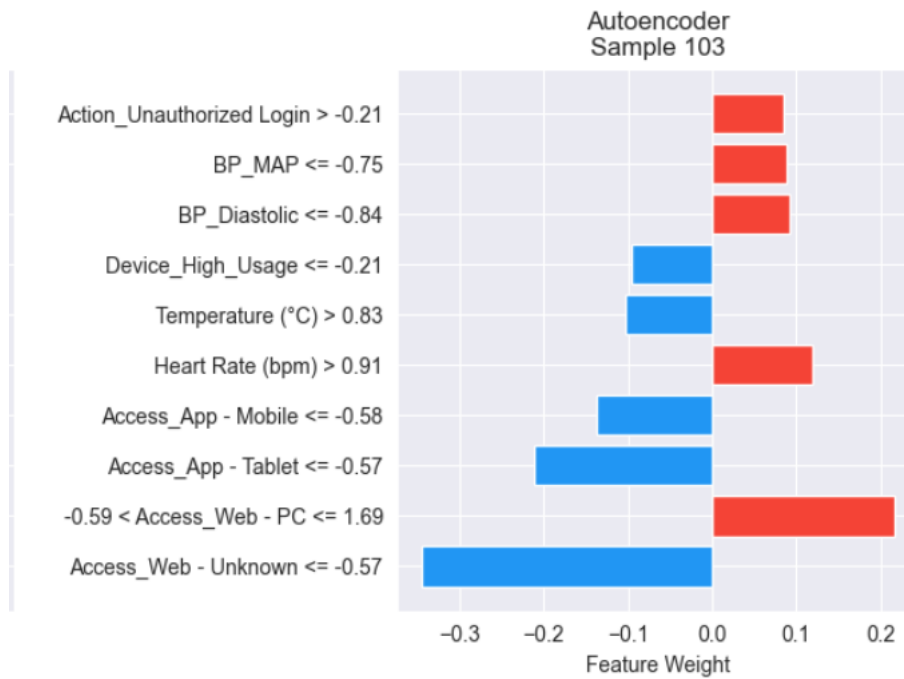


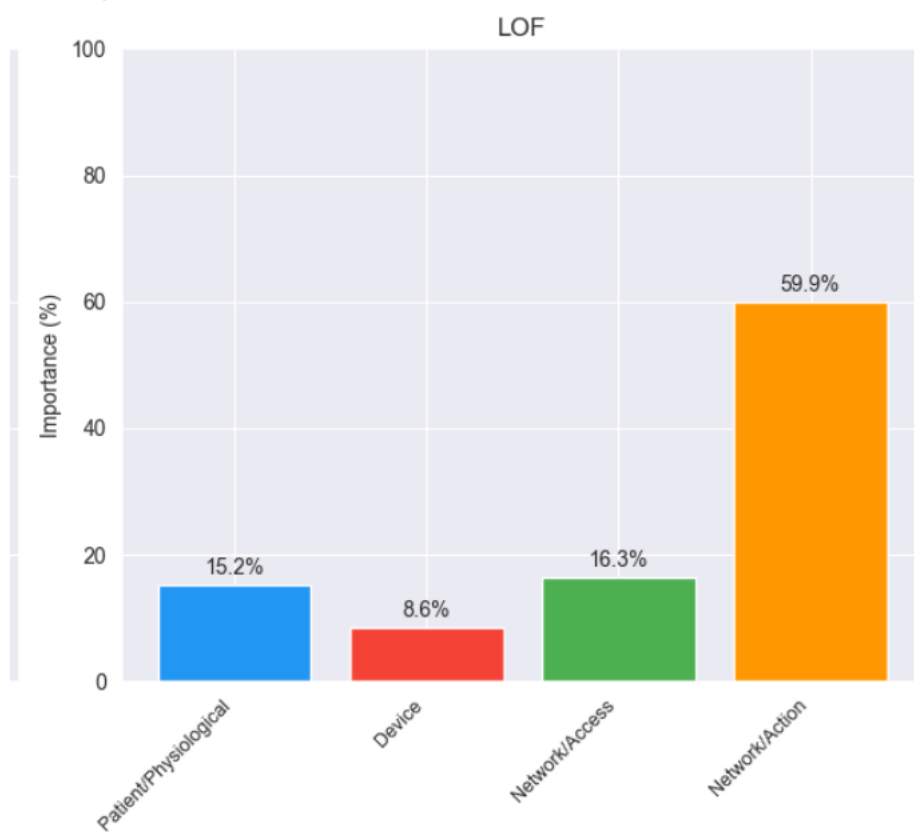
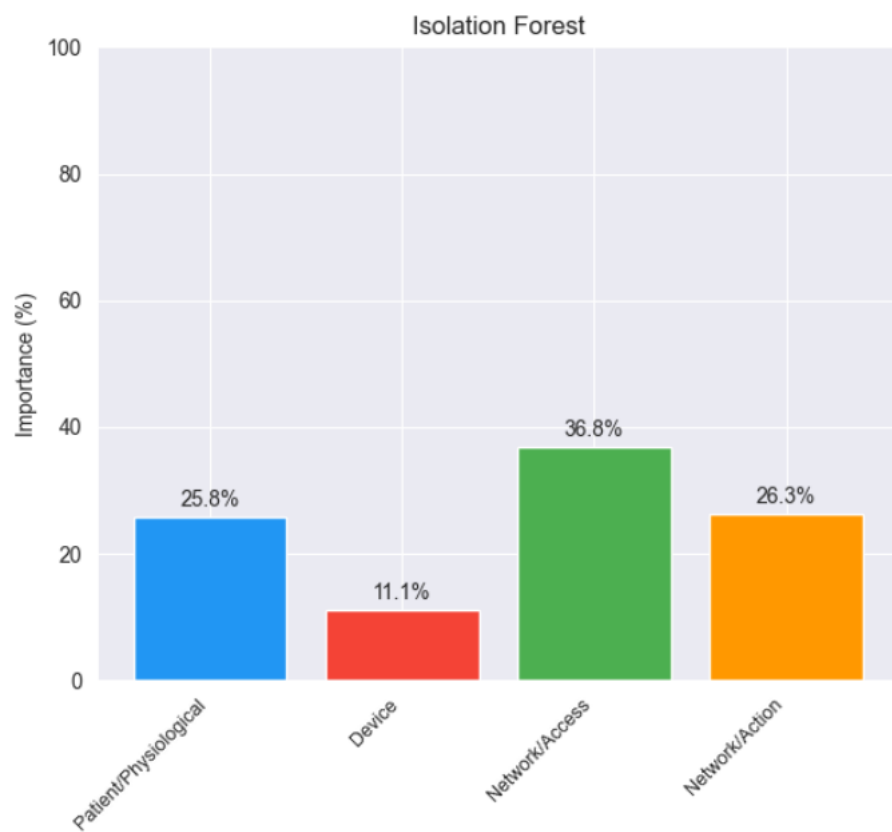


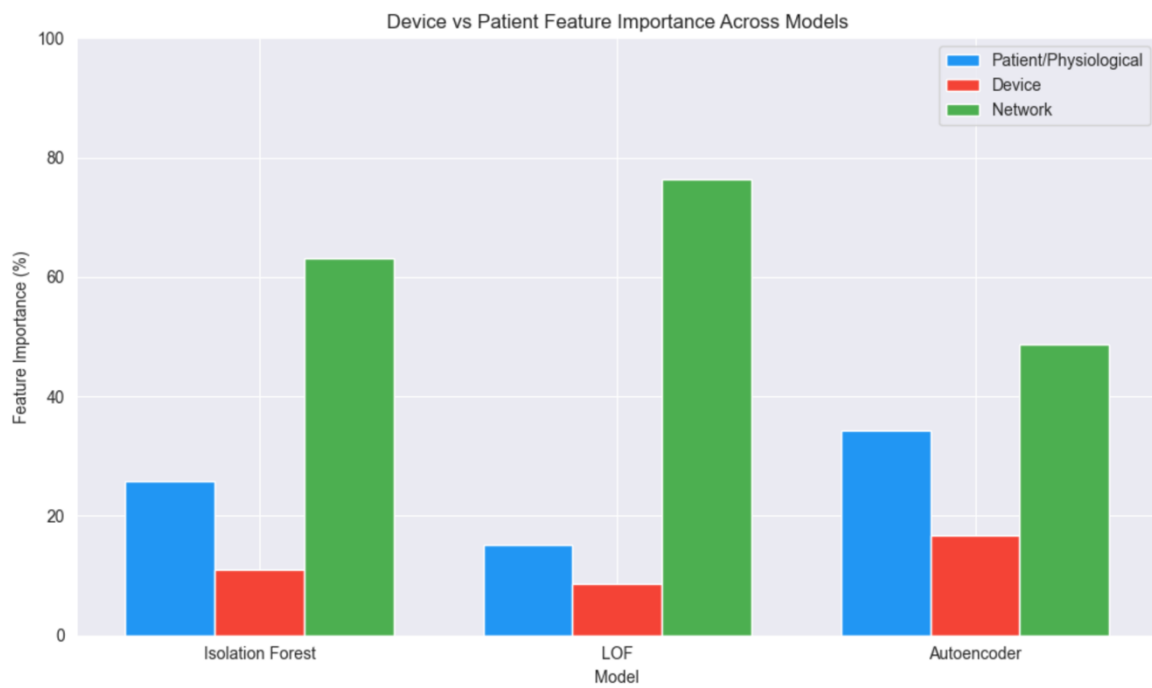
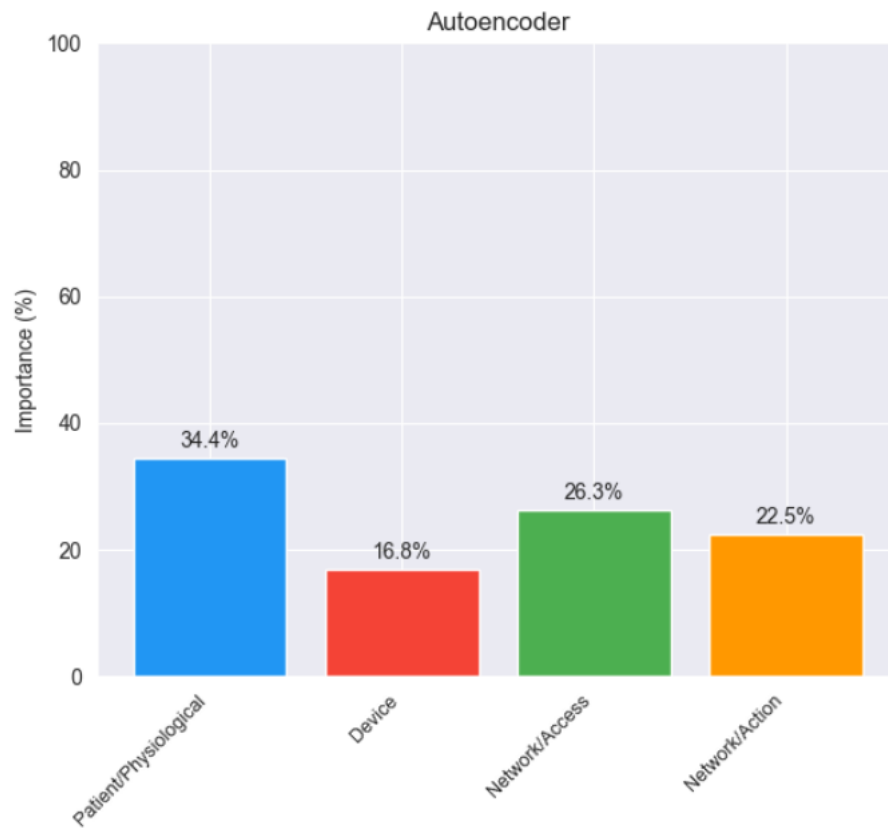


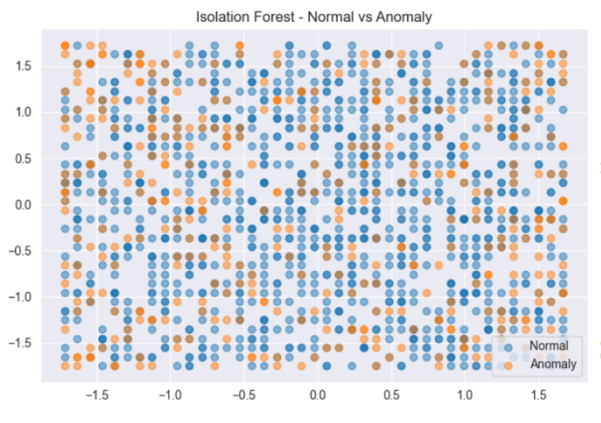
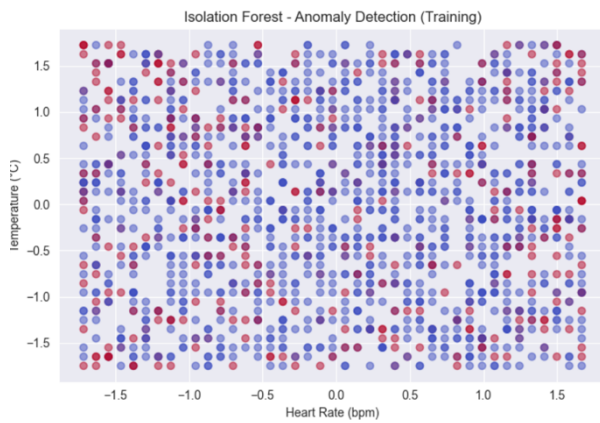
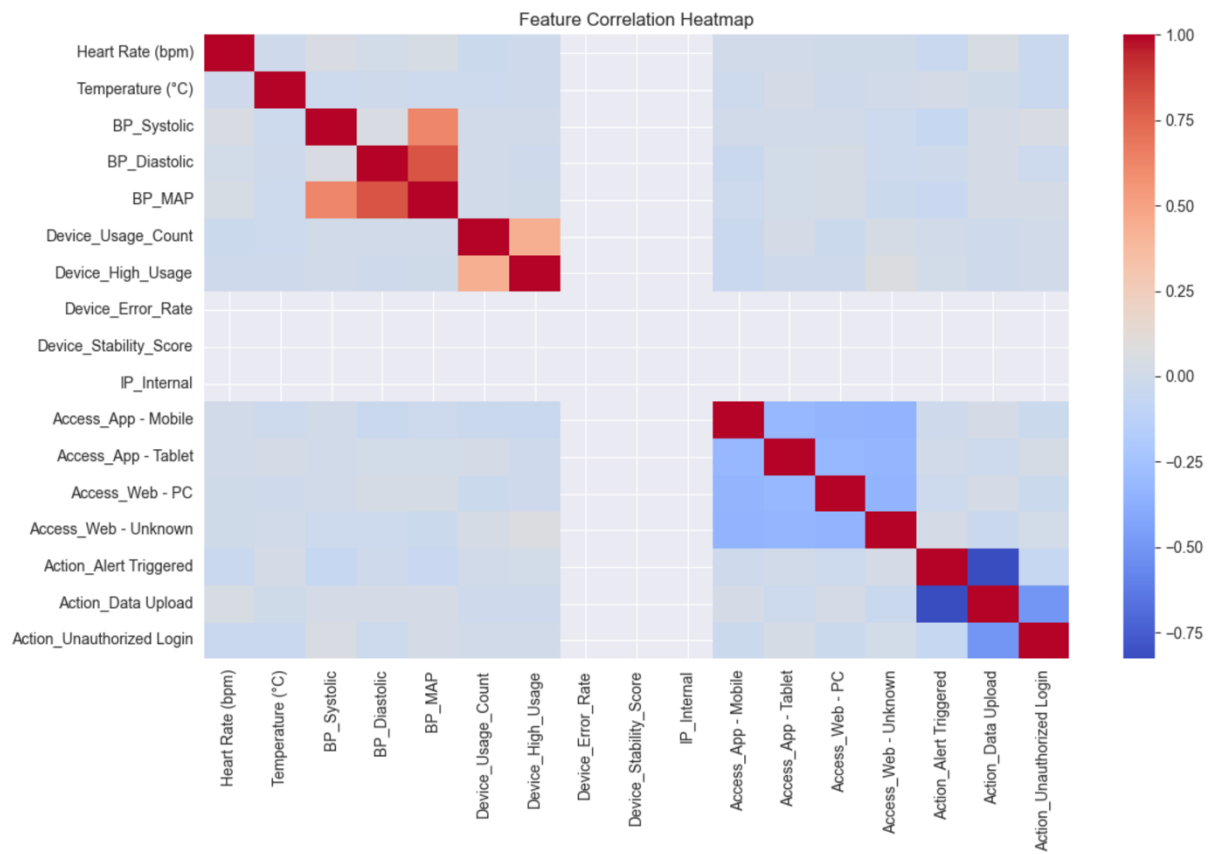






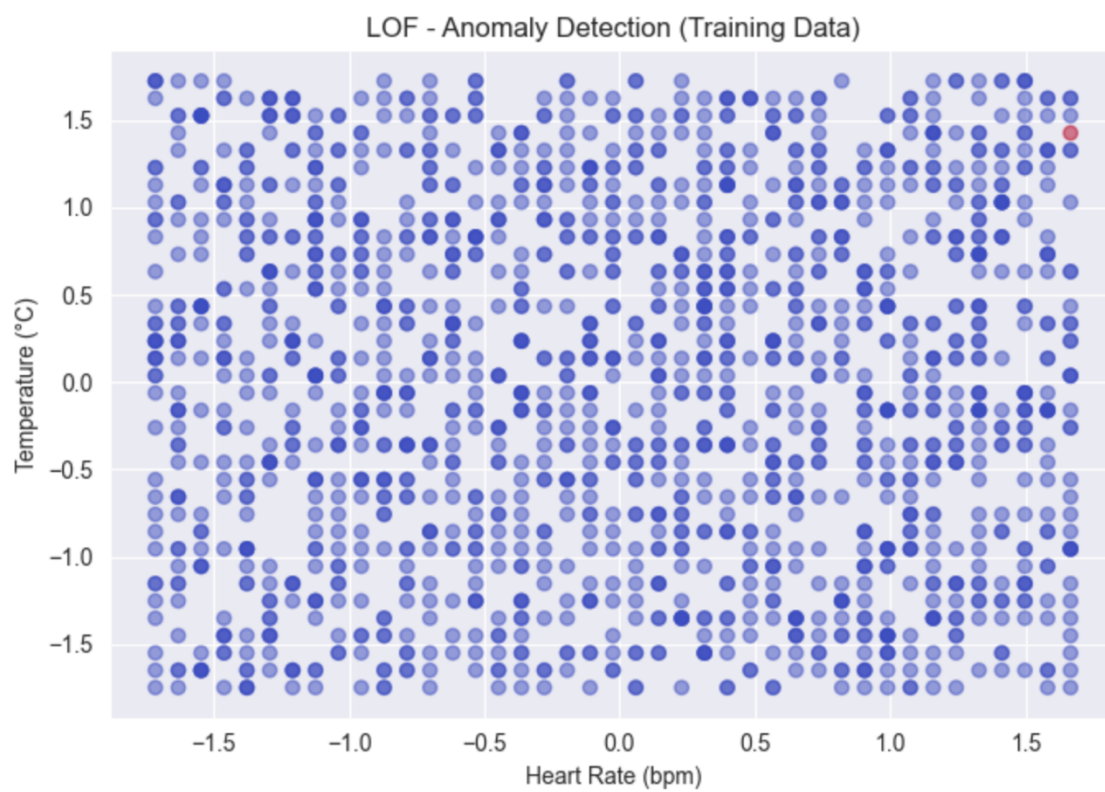
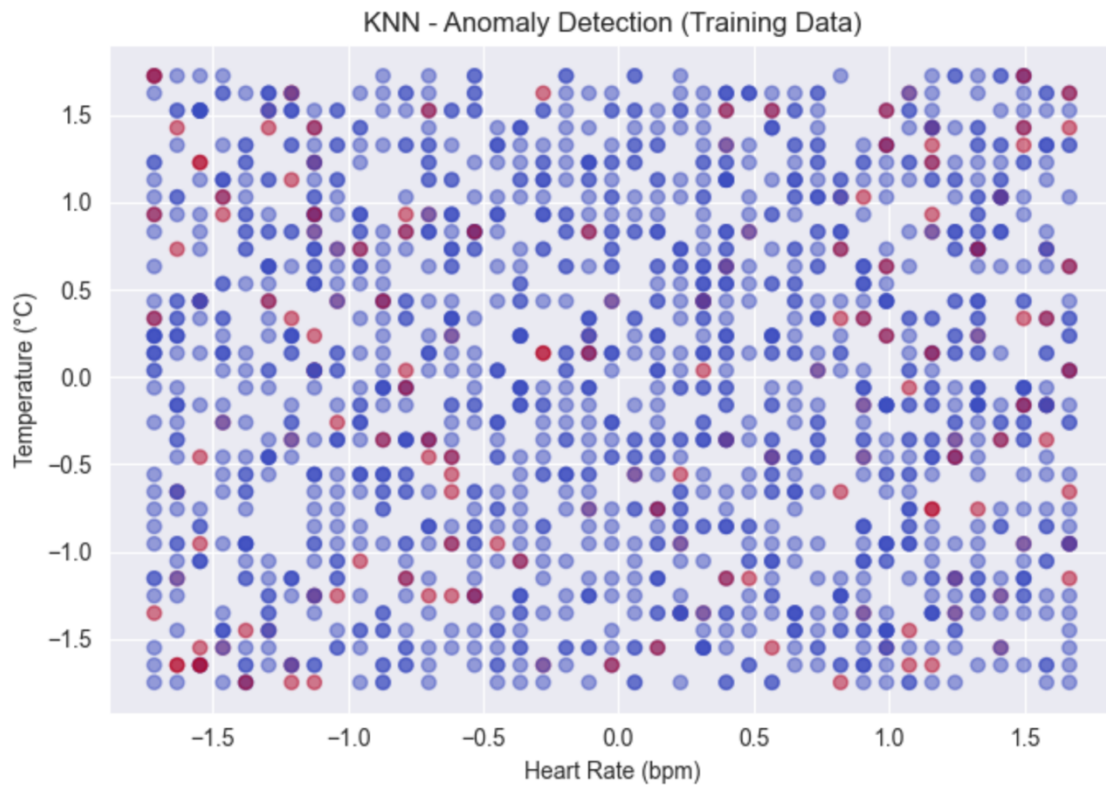






KNN - Training anomalies detected: 160/1600

KNN threshold: 1.4585



LOF - Training anomalies detected: 1/1600